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Pattaraphon Kenny Wongchamcharoen

Contact Information

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Education

University of California, Berkeley — Expected May 2027
B.S. Industrial Engineering & Operations Research
Minors in Data Science and Mathematics
GPA: 3.97/4.00

Selected Coursework

Introduction to Analysis (A), Nonlinear & Discrete Optimization (A+), Optimization Modeling (A+), Linear Programming & Network Flows (A), Introduction to Stochastic Processes (A+), Probability and Risk Analysis (A+), Engineering Statistics (A), Machine Learning & Data Analytics I (A+), II (A), Abstract Linear Algebra (A-)
Doctoral (In Progress): Mathematical Programming I, Applied Stochastic Process I

Research Interests

Methods: Stochastic Modeling and Queueing Theory; Optimization; Game Theory; Statistical and Empirical Methods; Applied Machine Learning

Applications: Human–AI Operations and AI Reliability; Service and Transportation Systems, Public-Sector and Platform Operations

Publications & Working Papers

[1] Do Large Language Models (LLMs) Understand Chronology?
Wongchamcharoen, P. K., & Glasserman, P.
Proceedings of the 2026 AAAI Conference on Artificial Intelligence (AAAI-26)
Oral – Student Abstract & Poster Program (Top 11%)
Poster – *AI4TS: AI for Time Series Analysis (AAAI-26 Workshop)*
[*proceedings*] [*arXiv*] [*code*]

[2] CentaurBench: Benchmarking LLM Capabilities on Augmenting vs. Automating Real-World Work Tasks
Wongchamcharoen, P. K., Gulati, K., Fong, M. M., & Nagaraj, A.
NBER Working Paper #35663
Poster – *NeurIPS 2026 Workshop on Agentic AI Benchmarks and Applications for Enterprise Tasks (AABA4ET)*
[*nber*] [*arXiv*] [*project dashboard*]

[3] On-Off Systems with Strategic Customers
with Y. Sun, Z. Liu, and C. Yan
Revise and Resubmit (R&R) at Operations Research
[*ssrn*]

Other Research Projects

[1] Data-Driven Evaluation of Board of Directors Effectiveness
Wongchamcharoen, P. K., Stringer, C., Hwang, B., Liu, I., & Li, K.
Best Data Visualization, Berkeley CDSS Data Discovery Symposium 2025
[*poster*]

[2] Mixed-Integer Linear Program for Options Pricing and Portfolio Optimization
Bin Abdulla, Q. M., **Wongchamcharoen, P. K.**, Jamari, A., & Lee, J.
1st Runner-up, Wells Fargo & IEOR Bay Area Decision Sciences Summit 2025
[*poster*] [*code*]

[3] Calyber: A Ridesharing Game
Shen, Y., Yan, C., & Yan, J.
INFORMS Transactions on Education, 26(3):218–225, 2026
Research assistant contribution; Runner-up, 2025 INFORMS Case Competition
[*article*] [*award*]

Talks & Presentations

"On-Off Systems with Strategic Customers." (with Y. Sun, Z. Liu, and C. Yan). Presentation at the *2026 INFORMS Annual Meeting, MSOM Invited Session* in San Francisco, CA. [*slides*]

Wongchamcharoen, P. K., Gulati, K., Fong, M. M., and Nagaraj, A. (2026). "CentaurBench: Benchmarking LLM Capabilities on Augmenting vs. Automating Real-World Work Tasks." Presentation at the *2026 Wharton Generative AI & Business Conference* in San Francisco, CA. [*slides*]

Wongchamcharoen, P. K. (2026). "AI Is Getting Smarter... Are We Cooked?: Research Perspectives on the Capabilities, Limits, and Future of AI." Invited talk at the *Berkeley Engineering Summer Machine-Learning & AI Research Training (BeSMART) Program* in Berkeley, CA. [*slides*]

Wongchamcharoen, P. K. and Glasserman, P. (2026). "Do LLMs Understand Chronology?" Presentation at the *2026 IISE Annual Conference* in Arlington, TX.

Wongchamcharoen, P. K. and Glasserman, P. (2026). "The Temporal Blind Spots of AI: Implications for IEOR Applications." Presentation at the *Berkeley IEOR Annual Advisory Board Meeting* in Berkeley, CA. [*press*]

Wongchamcharoen, P. K. and Glasserman, P. (2026). "Do LLMs Understand Chronology?" Oral presentation at the *40th Annual AAAI Conference on Artificial Intelligence* in Singapore. [*poster*] [*talk*]

Research Experience

UC Berkeley Industrial Engineering & Operations Research Berkeley, CA
Research Assistant
Advisor: Prof. Chiwei Yan Jan 2024 – Present

- Extended the on-off dynamic queueing model with strategic customers to multi-server systems; formulated the throughput problem as a linear program and proved results on throughput scaling, demand saturation, and optimal service allocation under exogenous on-off policies, including a closed-form linear-scaling threshold in the homogeneous case via an explicit dual certificate.
- Developed a discrete-event simulator, learning dynamics, and equilibrium-computation algorithms for the stochastic system, numerically demonstrating convergence of throughput to the fluid limit in the exogenous setting; used a hierarchical Bayesian mixed-effects model on BART GTFS-RT and OD data to estimate rider patience and recover demand lost to long waiting times: 3.30% of total latent demand overall (2.47% on weekdays; 7.85% on weekends). Optimized dwell-time policies across 24 Yellow Line stations via sample-average approximation.
- Created case teaching materials for *Calyber*, a ridesharing simulation game, and deployed it in a graduate supply-chain course; built AI agents to benchmark profit and efficiency against student teams.

UC Berkeley Haas School of Business Berkeley, CA
Research Assistant, Data Innovation and AI Lab (DIAL)
Advisor: Prof. Abhishek Nagaraj Jan 2024 – Present

- Led a four-person team developing *CentaurBench*, a benchmark comparing LLM capabilities in automating professional tasks and augmenting another model's work.
- Engineered a multi-provider asynchronous Python evaluation pipeline across six model providers and an LLM-as-judge harness using anonymized round-robin comparisons, randomized response ordering, micro-rubrics, and *leave-family-out* masking; validated 15,000+ scored comparisons using rank correlations, inter-judge agreement, Krippendorff's alpha, and sensitivity analysis; created a reproducibility repository and public dashboard auditing outputs, rankings, and rationales.

Columbia Business School, Decisions, Risk, and Operations New York, NY
Summer Research Fellow
Advisor: Prof. Paul Glasserman Jun 2025 – Aug 2025

- Selected as 1 of 15 interns from more than 1,000 applicants; built a reproducible Python pipeline using GPT-4.1, o3, Claude 3.7 Sonnet, and other frontier LLM APIs to evaluate their chronological reasoning through basic sorting, conditional sorting, and anachronism detection; developed a statistical framework to compare performances using Spearman's and Kendall's rank correlations across model families and task configurations.
- Published the findings in the AAAI-26 proceedings; cited in OpenAI's 2026 working paper, "Scaling social science research" [*working paper*].

UC Berkeley School of Law Berkeley, CA
Student Researcher
Advisor: Prof. Frank Partnoy Jan 2025 – May 2025

UC Berkeley Haas School of Business Berkeley, CA
Research Assistant
Advisor: Prof. David Evan Harris Aug 2023 – Jan 2024

Professional Experience

Beats by Dre, Data Analytics Extern, Culver City, CA Jun–Aug 2024
Xcube Capital, Product Development & Analytics Intern, Singapore May–Aug 2024
MENTOR Global Consultants, Technical Product Mgmt. Intern Jan–Apr 2024
GUESS, Inc., Strategy Contract Consultant Aug–Dec 2023

Honors and Awards

Tau Beta Pi Scholarship (Tau Beta Pi Association) 2026
Bonder Foundation Scholarship (INFORMS) 2026
Dean's Lists (UC Berkeley College of Engineering) 2023–2026
CBS Research Internship Grant (Columbia Business School) 2025
Best Poster: Data Visualization Award (UC Berkeley CDSS) 2025
1st Runner-up, Bay Area Decision Sciences Summit (Wells Fargo) 2025
Cal Alumni Leadership Award Finalist (Cal Alumni Association) 2024
College of Engineering Mentorship Grant (UC Berkeley) 2024
ASEAN Scholarship (Singapore Ministry of Education) 2018

Teaching Experience

Tutor & Grader, University of California, Berkeley — Fall 2024–Present

- IEOR 145: Fundamentals of Revenue Management (Prof. Rajan Udwani)
- IEOR 142A: Machine Learning & Data Analytics I (Prof. Phillip Kerger)
- IEOR 172: Probability & Risk Analysis for Engineers; IEOR 120: Principles of Engineering Economics (Prof. Daniel Pirutinsky)
- CS 88: Computational Structures in Data Science (Prof. John DeNero)
- Tutor for lower- and upper-division courses in mathematics, data science, computer science, and IEOR, Berkeley's Center for Access to Engineering Excellence (CAEE)

Primary Lecturer, Berkeley Engineering GLOBE Center — Fall 2025–Present

- Developed and taught *Foundations of Machine Learning*, an introductory machine-learning course for underserved Bay Area high-school students (Program Advisor: Prof. Phillip Kerger). [*course website*]

Teaching Assistant, University of California, Berkeley — Spring–Summer 2026

- IEOR C253: Supply Chain & Logistics Management (Prof. Chiwei Yan)
- Berkeley Summer Machine Learning & AI Research Training Program (BeSMART); mentored student machine learning research projects and presentations, created assignments, and guest lectured on coding and agentic AI (Prof. Phillip Kerger)

Service and Leadership

Reviewer, NeurIPS 2026 Workshop AABA4ET, 2026

Institute of Industrial & Systems Engineers at Berkeley
Senior Advisor — Fall 2025 – Present
Director of Programs and Events — Fall 2023 – Spring 2024

Singapore Students Association at Berkeley
Vice President of External Relations — Spring 2024 – Fall 2024

Professional Memberships

Institute for Operations Research and the Management Sciences (INFORMS); Institute of Industrial & Systems Engineers (IISE); Tau Beta Pi Engineering Honor Society

Skills & Languages

Python (NumPy, Pandas, Matplotlib, scikit-learn, Pyomo, TensorFlow); Java; SQL; Apache Spark; LLM APIs (OpenAI, Anthropic, Google Gemini); HuggingFace; LaTeX

English (Fluent); Thai (Native); Mandarin Chinese (Elementary)